

## Element D

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Our solutions are based on a number of factors including price, how much material we will be using, and characteristics required by our stakeholder, Dr. Brossman. In addition, our solutions are based on a number of qualities of design that we wanted in our respiratory device. Below is attached a morphological chart to show all our concepts. Below, there is also a weighted decision matrix that accounts for each design quality with each concept, however, this is meant to help us with the final decision.

### Morphological Chart

| Function           | Concept 1                | Concept 2                     | Concept 3            | Concept 4        | Concept 5     |
|--------------------|--------------------------|-------------------------------|----------------------|------------------|---------------|
| Display Pressure   | Light Meter              | Dial (Manometer)              | Digital Meter        | Sensor           | Barometer     |
| Seal in air        | One smooth piece         | Sealed with glue              | Melt pieces together | Welding          | Rubber Gasket |
| Restrict airflow   | Spring                   | Tighten tube                  | Water/liquid         | Elastic Membrane |               |
| Adjust restriction | Mechanical dial adjuster | Arduino running a servo motor | Pressure Canister    |                  |               |
| Store Data         | Second dial              | Display Screen                | Light Strip          | USB Connection   |               |

### Decision Matrix

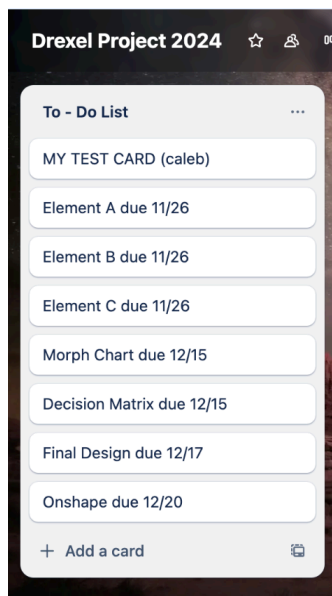
|                     |        | Design 2 |       | Design 3 |       | Design 4 |       |
|---------------------|--------|----------|-------|----------|-------|----------|-------|
| Qualities of Design | Weight | Rating   | Value | Rating   | Value | Rating   | Value |
| Display pressure    | 10%    | 3.0      | 0.30  | 2.0      | 0.20  | 2.5      | 0.25  |
| seal in air         | 30%    | 4.0      | 1.20  | 4.0      | 1.20  | 3.0      | 0.90  |

|                    |      |     |      |     |      |     |      |
|--------------------|------|-----|------|-----|------|-----|------|
| restrict airflow   | 30%  | 1.0 | 0.30 | 1.0 | 0.30 | 0   | 0.00 |
| adjust restriction | 30%  | 2.0 | 0.60 | 1.0 | 0.30 | 2.0 | 0.60 |
|                    | 100% |     | 2.40 |     | 2.00 |     | 1.75 |

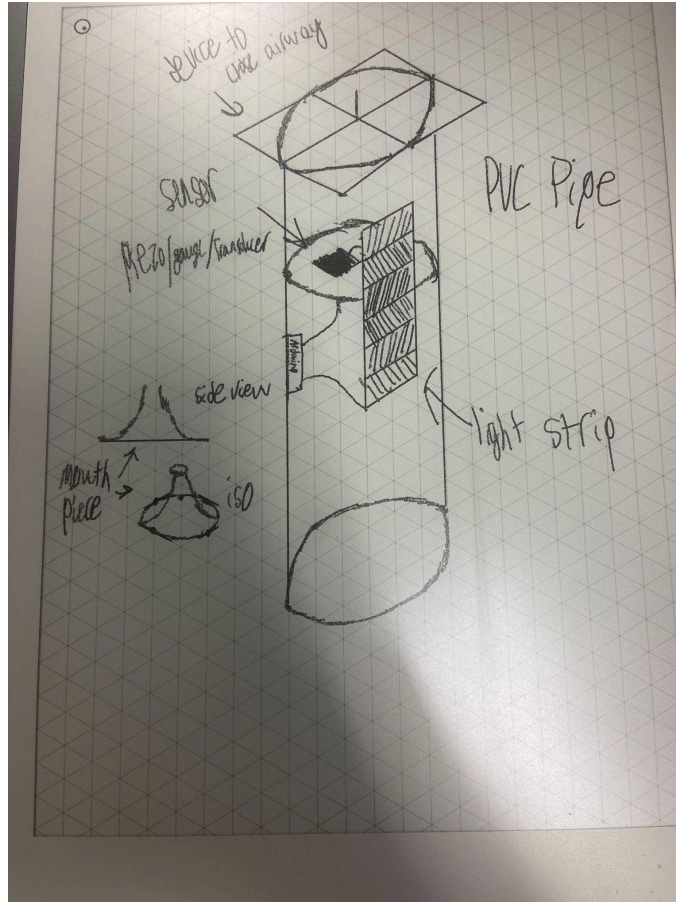
Our rationale:

After reading Dr. Brossman's email thoroughly and talking with real engineers, our group settled on the design that cost the least. The prototype includes a PVC pipe, two 3D-printed parts, a bucket of water, and a mouthpiece. The goal of this project is to create a simple barometer. The negative pressure causes the water to rise, signaling how strong one inhales. The actual MIP respiratory device usually uses a dial.

We decided that we were going to do Design 2 for our device. This is because it is the most efficient design to build. The design is able to seal in the air the best, although it is tied with Design 3, Design 2 is stronger in adjusting the resistance which was a top priority of ours. Design 2 overall was also easy to build so it was time-efficient. This means that we will be able to have more time to calibrate the prototype and overall produce a better product.

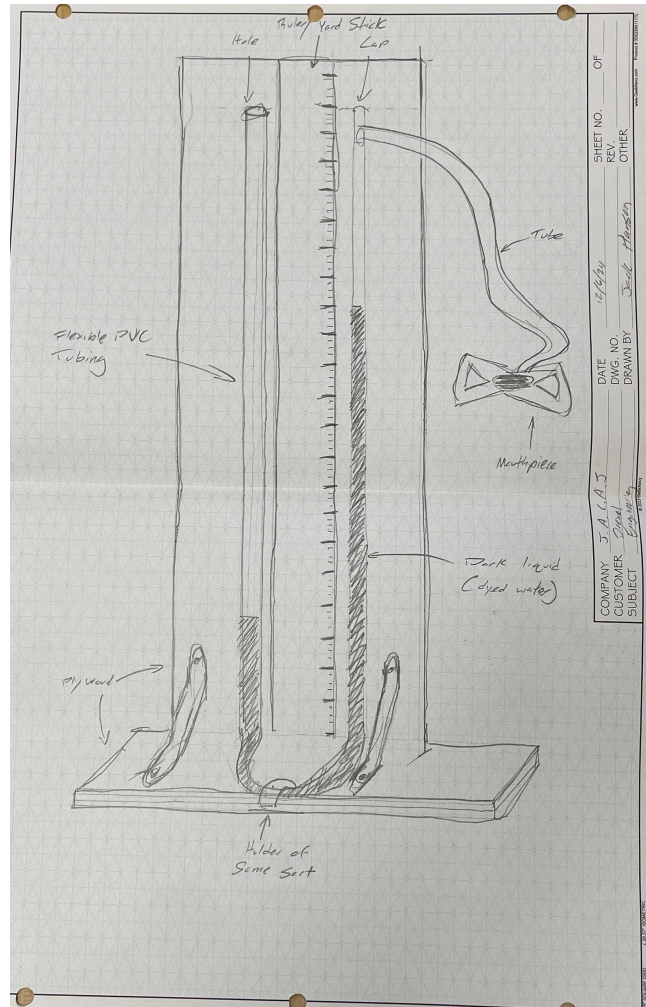


On the left is an image of our to-do list. This list helped us keep track of tasks we need to complete. In addition, Trello keeps track of important dates to make sure our project is done on time and done efficiently.



### **Design 1**

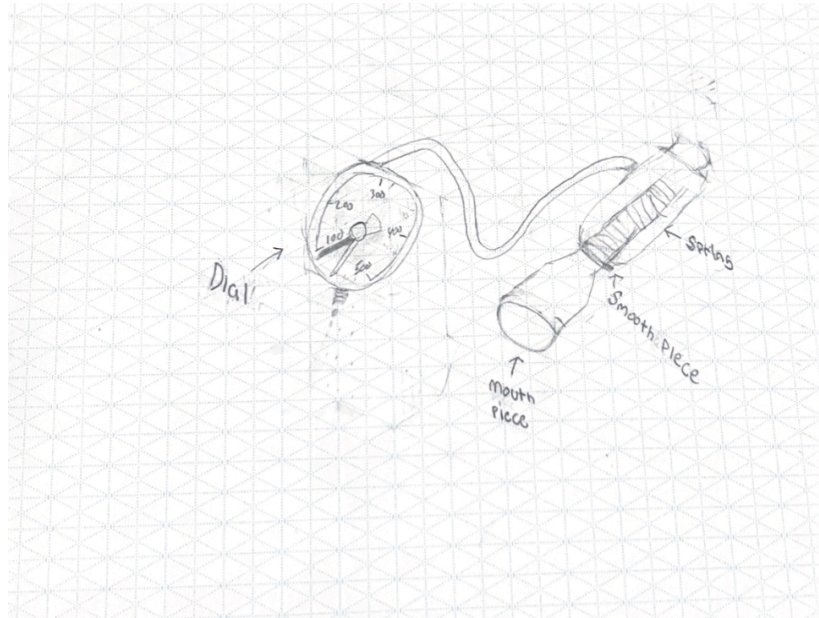
The cost of this design is in the middle of the road. The idea is to have a sensor inside a tube with one end closed off to read changes in pressure. The mouthpiece can be removed to allow adjustments to the sensor. The issue is that the electronics are complicated, as is keeping them separate from any fluid that could come into contact with them. Overall, it's probably a bit too complicated to work as a design option for a high school group.



## **Design 2**

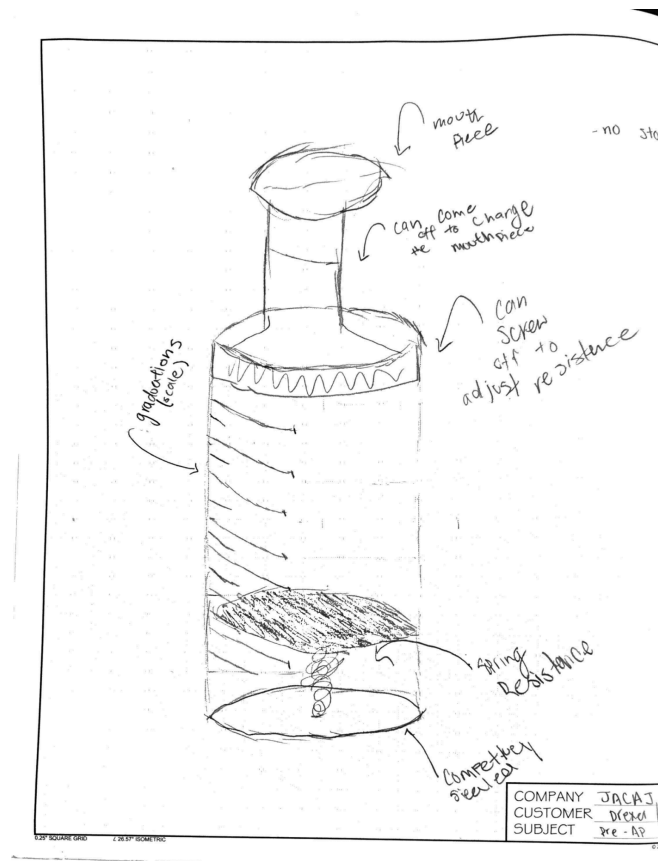
The conceptual design above should be relatively simple to build. Below is a video which the idea derives from. When the student inhales through the tube, the liquid rises in response to the pressure. Advantages to this design is that the idea of the project is extremely inexpensive and easy to make. However, some disadvantages are possible concerns with the liquid. The diameter of the PVC tube needs to be big enough so the subject does not make contact with the liquid.

[Link to video](#)



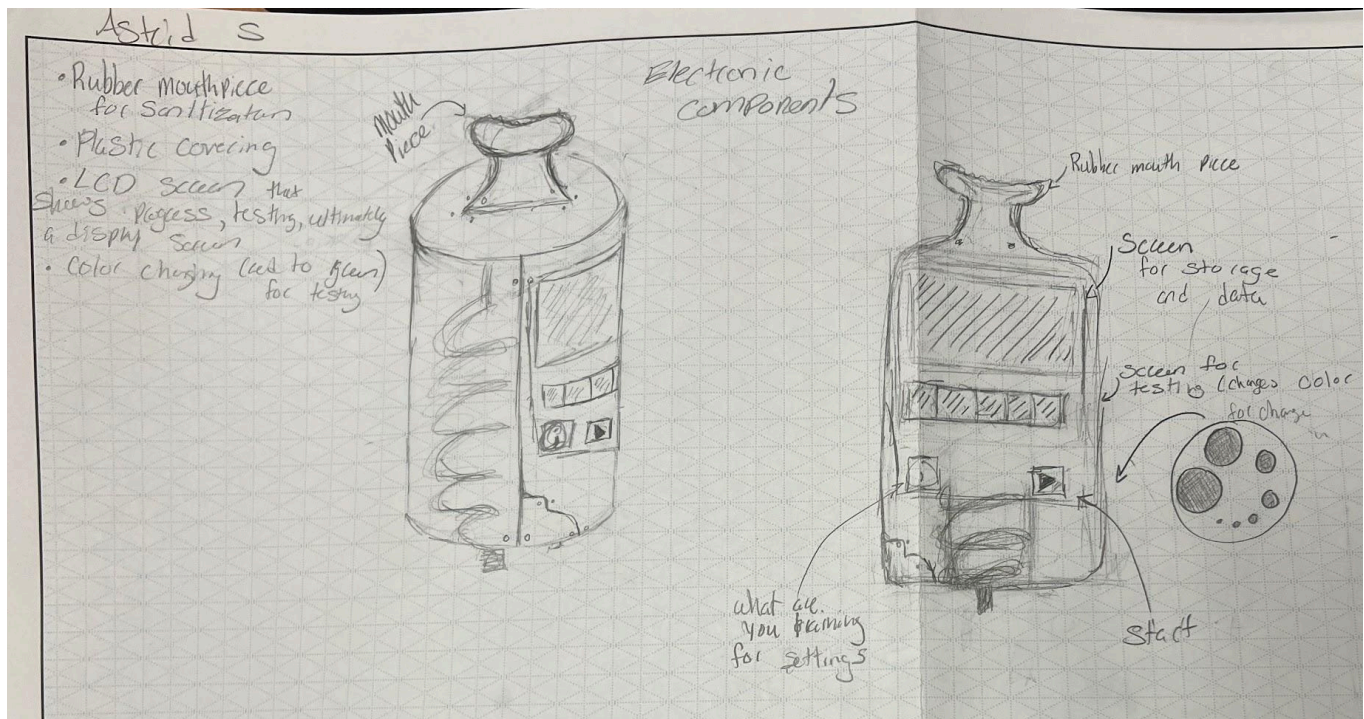
### **Design 3**

Design 2 uses a spring mechanism to measure the pressure and uses a dial to display the pressure. There will be two separate pieces, which are the mouthpiece and a plastic chamber to hold the spring mechanism. This is a simple and inexpensive design that is not only easy to use but also easy to make. The advantages of this design is that it is straightforward, small, and has adjustable resistance. In addition, there is visual feedback which is a very unique characteristic, it is cost effective, and eco-friendly. However, some disadvantages of this design is that it isn't very accurate, precise, and doesn't provide the most amount of feedback as compared to other IMT devices. This isn't that big an issue though since it is more for educational use rather than for medical use.



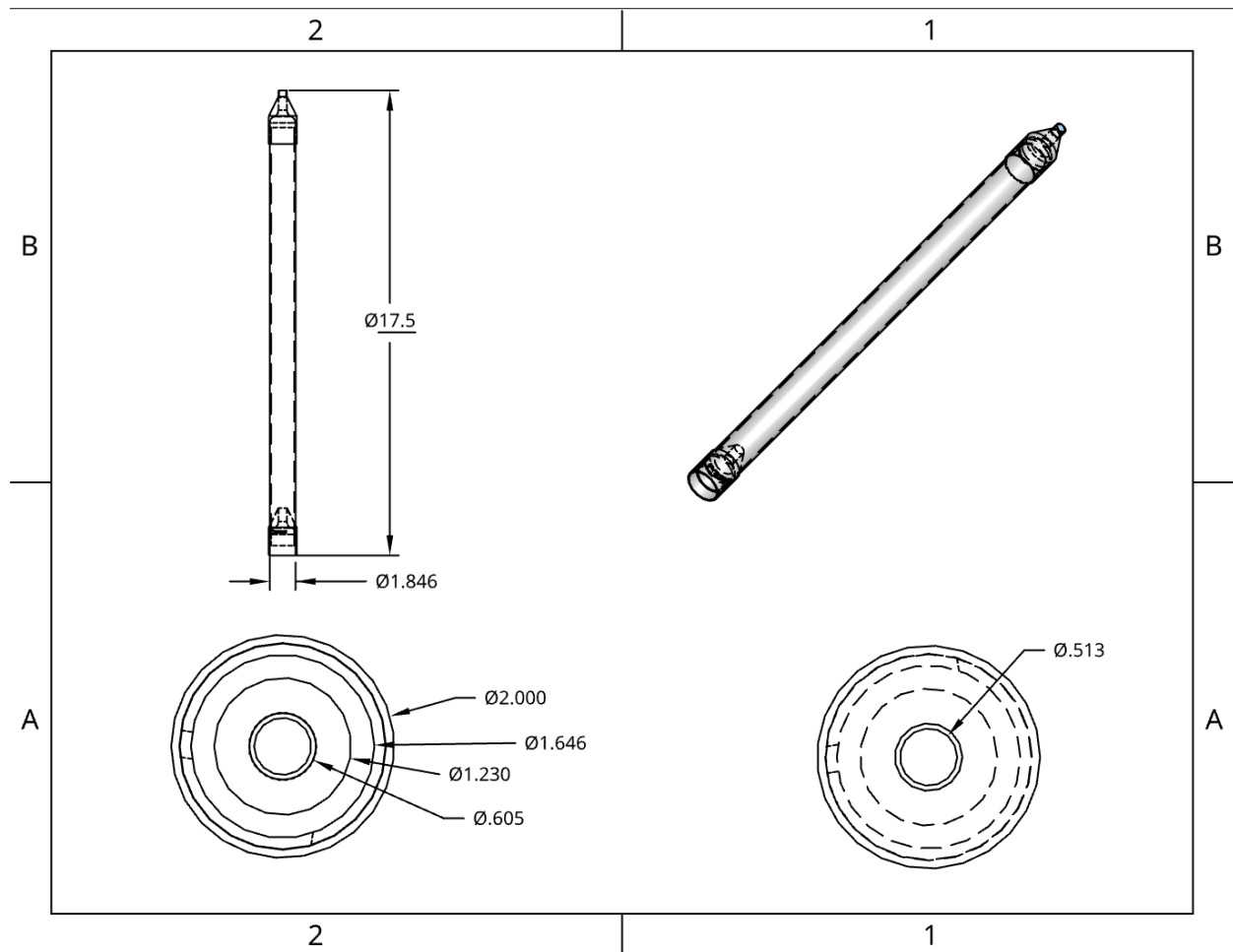
#### Design 4

This design's main purpose is to be cheap and educational rather than it being pricey and accurate. The way this device works is similar to the Philips Threshold IMT where the user breathes in through a mouthpiece and there is adjustable resistance used by a spring. The mouthpiece comes off so each student or every few students can have one. The top part can screw off to adjust the resistance of the spring. The good aspects about this design is that it is hygienic because each person or every few people can get their own mouthpieces. Next there is adjustable resistance which is an aspect that a lot of patients enjoy with their Philips Threshold IMT. Finally, the device won't take long to make and is super cheap compared to the other designs. However, there are some downsides to this design. First being that it is not the most accurate design. Meaning that the data shown might not be accurate but this isn't a big issue as one of this design's main purposes is to be educational. This lets the students truly understand how these types of devices work and let them try it out for themselves. To end off, the data cannot be stored in the device. This isn't a big issue at all as Dr. Brossman states as data storage isn't a priority



### Design 5

If our group had unlimited resources, time, and money, this would be the ideal device. The casing of the model is made out of hard plastic and the mouthpiece is rubber. The detachable rubber mouthpiece helps to keep the user safe from any bacteria, due to the simple cleaning mechanism. What makes this device the most expensive/ complex MIP is the electronic components. There are two screens, a large LCD screen and a color changing meter. The LCD screen is used as a display screen for testing, storage, and to show progress. The color changing meter (Red to Green) is used during live testing to show your inhalation. The (i) is used to change the settings. The circle drawing on the right is inside of the device. Changing the setting will change the hole size, making it easier/ harder for the user to inhale.



### **Design Prototype as of January 13, 2025**

The image above shows a CAD drawing of our final prototype. We still went with the idea of Design 2 but shrunk it down a bit more and made it less bulky. At first, we realised that the initial idea was going to be way too big and too complicated. We ended up deciding on a prototype that was shaped more like a pencil that will be in a bucket of water. We decided to use a clear PVC pipe and 3D printed pieces on top and bottom of the pipe. The top piece is where the user will put their mouth and breath in, and the bottom piece will be how the user adjusts the resistance.